# Supplementary material for *Dynamic consequences of declining vaccination coverage in a heterogeneous world*

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[**Emily:** *I've rewritten the first half of the first section to define the model equations and derive the equilibria explicitly.*]

## 1 Theoretical honeymoon time in the SIR model

We consider here how reductions in vaccination can lead to rebounds in disease prevalence. In particular, we will focus on the post-vaccination "honeymoon", or how long it takes for the disease to rebound after vaccination starts declining. We use the term "effective immunization coverage" throughout for the proportion successfully immunized, thereby accounting for both vaccination coverage and imperfect vaccine efficacy.

We begin with an SIR (Susceptible, Infected, Recovered) model of disease transmission. We assume that births and deaths occur at the same rate, $\mu$, and some proportion, $v$, of births are effectively immunized. We further assume that new infections occur upon contact between susceptible and infected individuals at rate $\beta$, and infected individuals recover at rate $\gamma$. The model can be defined by the following system of ordinary differential equations:

$$\frac{dS}{dt} = \mu(1-v) - \beta SI - \mu S \tag{1}$$

$$\frac{dI}{dt} = \beta SI - (\gamma+\mu)I \tag{2}$$

$$\frac{dR}{dt} = \mu v + \gamma I - \mu R \tag{3}$$

where $S$, $I$, and $R$ represent the proportions of the population that are susceptible, infected, and recovered, respectively.

One way to understand the dynamics of the SIR model is through its equilibrium behavior, or the conditions under which the system does not change over time. The equilibrium conditions are given by setting the derivatives to zero, and for the $I$ compartment, we have

$$\frac{dI}{dt} = \beta SI - (\gamma+\mu)I = I(\beta S - (\gamma+\mu)) = 0, \tag{4}$$

which demonstrates the existence of two equilibria:

1. disease-free equilibrium, where $I^* = 0$ and $S^* = 1 - v$, and
2. endemic equilibrium, where $I^* = \frac{\mu}{(\gamma+\mu)}\big((1-v) - 1/\mathcal{R}_0\big) > 0$ and $S^* = \frac{\gamma+\mu}{\beta} = 1/\mathcal{R}_0$. This equilibrium point only exists and is positive when $v < 1 - 1/\mathcal{R}_0$. [**Emily:** *the exact equation for $I^*$ in this case is less relevant for our purposes, but I've included it here for completeness*]

Within this framework, we imagine a population with some "pre-decline" effective immunization coverage, $v_0$, that experiences an instantaneous drop in coverage to $v$. We consider only the disease-free case in which the pre-decline effective immunization coverage is above the herd-immunity threshold, $v_0 > 1 - 1/\mathcal{R}_0$, equivalently $S^* = 1 - v_0 < 1/\mathcal{R}_0$; otherwise there is no positive honeymoon period. [**Emily:** *flagging that we haven't formally introduced the definition of honeymoon period yet, but perhaps safe to say the reader has been through the main text and has the general idea*]

We hypothesize that the duration of the honeymoon period, or informally, the time until the first rebound outbreak, will depend on a number of factors including the basic reproduction number of the pathogen $\mathcal{R}_0$, the equilibrium proportion susceptible $S^*$, the birth rate $\mu$, the post-decline effective immunization coverage $v$, and the timing of importations. Because the timing and establishment of importations cannot be predicted by this deterministic model, we define the theoretical honeymoon time as the time until $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 S(t)$ crosses 1. This provides a lower bound on the actual rebound time, because rebound additionally requires an importation to arrive and establish.

We can calculate analytically the time at which $\mathcal{R}_{\text{eff}}$ crosses 1 under the assumptions of local extinction ($I = 0$), constant post-decline effective immunization coverage $v$, and a constant population scaled to 1. The proportion susceptible satisfies

$$\frac{dS}{dt} = \mu(1-v) - \mu S = \mu\big[(1-v) - S\big]. \tag{5}$$

Because the population is initially at the disease-free equilibrium associated with the pre-decline effective immunization coverage $v_0$, we have $S(0) = 1 - v_0$. Since $v < v_0$, we have $S(0) < 1 - v$, so Equation (5) shows that $S(t)$ increases toward $1 - v$. Thus we have a first-order linear initial-value problem, whose solution is

$$S(t) = 1 - v - (v_0 - v)e^{-\mu t}. \tag{6}$$

The critical effective immunization coverage (corresponding to the herd-immunity threshold) is

$$v_{\text{crit}} = 1 - \frac{1}{\mathcal{R}_0}. \tag{7}$$

If effective immunization coverage declines from above to below this level, so that

$$v < v_{\text{crit}} < v_0, \tag{8}$$

then $S(t)$ increases from $1 - v_0$ and crosses $1/\mathcal{R}_0 = 1 - v_{\text{crit}}$ after a finite time. Setting $\mathcal{R}_{\text{eff}}(t) = \mathcal{R}_0 S(t) = 1$ gives the theoretical honeymoon time,

$$T = \frac{1}{\mu}\log\left(\frac{v_0 - v}{v_{\text{crit}} - v}\right) = \frac{1}{\mu}\log\left(\frac{v_0 - v}{1 - 1/\mathcal{R}_0 - v}\right). \tag{9}$$

This expression is positive because $v < v_{\text{crit}} < v_0$ implies $(v_0 - v)/(v_{\text{crit}} - v) > 1$. If $v \geq v_{\text{crit}}$, $\mathcal{R}_{\text{eff}}$ never exceeds 1 and the theoretical honeymoon time is infinite; if $v_0 \leq v_{\text{crit}}$, the population is not initially above the herd-immunity threshold and there is no positive honeymoon period of this kind.

Equivalently, the pre- and post-decline values of effective immunization coverage that yield the same honeymoon time $T$ satisfy

$$v_0 = v + e^{\mu T}(v_{\text{crit}} - v) = v\big(1 - e^{\mu T}\big) + e^{\mu T} v_{\text{crit}}. \tag{10}$$

For fixed $T$ and $\mu$, Equation (10) defines a straight line in the plane whose horizontal and vertical coordinates are the post- and pre-decline effective immunization coverages, $v$ and $v_0$, respectively. The line has slope $1 - e^{\mu T} < 0$ and vertical intercept $e^{\mu T} v_{\text{crit}}$. Consequently, combinations of $v$ and $v_0$ that produce the same theoretical honeymoon time lie along straight lines. This explains the linear contours in the heatmap of theoretical honeymoon time in the main text.

[**David:** *Heatmap file:* `figures/honeymoon_SIR_with_linear_heatmap.pdf`*. I have referred to it descriptively because the main manuscript is not in LaTeX and its figure number may change.*]

## 2 Age-structured model

Our age-structured model defines a discrete set of $A$ age classes. Transmission between individuals of different age classes is governed by a who-acquires-infection-from-whom (WAIFW) matrix, $\theta_{i,j}$, where $i$ and $j$ index age classes. The full system of ordinary differential equations is as follows:

$$\begin{aligned}
\frac{dS_{i=1}}{dt} &= \mu(1-v) - \sum_{j=1}^{A} \theta_{1,j} S_1 I_j - \mu S_1 - \alpha_1 S_1 \\
\frac{dS_{i\neq 1}}{dt} &= \alpha_{i-1} S_{i-1} - \sum_{j=1}^{A} \theta_{i,j} S_i I_j - \mu S_i - \alpha_i S_i \\
\frac{dI_{i=1}}{dt} &= \sum_{j=1}^{A} \theta_{1,j} S_1 I_j - (\gamma + \mu) I_1 - \alpha_1 I_1 \\
\frac{dI_{i\neq 1}}{dt} &= \alpha_{i-1} I_{i-1} + \sum_{j=1}^{A} \theta_{i,j} S_i I_j - (\gamma + \mu) I_i - \alpha_i I_i \\
\frac{dR_{i=1}}{dt} &= \mu v + \gamma I_1 - \mu R_1 - \alpha_1 R_1 \\
\frac{dR_{i\neq 1}}{dt} &= \alpha_{i-1} R_{i-1} + \gamma I_i - \mu R_i - \alpha_i R_i
\end{aligned}$$

where $\alpha_i$ is the rate of aging out of age class $i$, $\mu$ is the birth and death rate, $\gamma$ is the recovery rate, and $v$ is the effective immunization coverage at birth. This model assumes a total population size of 1, and compartments represent the proportion of individuals in each age class.

We can calculate the basic reproduction number, $\mathcal{R}_0$, and the effective reproduction number, $\mathcal{R}_{\text{eff}}$, for this model using the Next Generation Matrix method. In this case, the next generation matrix, $K$, is given by

$$K = \frac{1}{\gamma+\mu}\begin{bmatrix}\theta_{1,1}S_1(t) & \theta_{1,2}S_1(t) & \cdots & \theta_{1,n}S_1(t)\\ \theta_{2,1}S_2(t) & \theta_{2,2}S_2(t) & \cdots & \theta_{2,n}S_2(t)\\ \vdots & \vdots & \ddots & \vdots \\ \theta_{n,1}S_n(t) & \theta_{n,2}S_n(t) & \cdots & \theta_{n,n}S_n(t)\end{bmatrix} = \frac{1}{\gamma+\mu}\theta\overrightarrow{S(t)},$$

for which we calculate the dominant eigenvalues numerically.

# 3 Unity beta calculation

To quantify the role of age-structured dynamics, we will use a theoretical quantity called "unity beta". Consider an age-structured transmission process that is governed by heterogeneous contacts between age groups, where the number of new infections in the age structured model is

$$\text{age-structured infections} = \frac{1}{N}\sum_{i=1}^{n}\sum_{j=1}^{n}\theta_{i,j}S_iI_j$$

where $\theta_{i,j}$ is the transmission rate between age groups $i$ and $j$. We can then calculate a time-varying quantity, $\hat{\beta}(t)$, called "unity beta" that ensures the number of new infections in the age-structured model is equal to the number of new infections in a non-age-structured model, where

$$\begin{aligned}\text{age-structured infections} &= \frac{1}{N}\sum_{i=1}^{n}\sum_{j=1}^{n}\theta_{i,j}S_iI_j\\ &= \frac{\hat{\beta}(t)}{N}\sum_{i=1}^{n}S_i\sum_{j=1}^{n}I_j\\ &= \text{homogenous infections}\end{aligned}$$

given that

$$\hat{\beta}(t) = \frac{\sum_{i=1}^{n}\sum_{j=1}^{n}\theta_{i,j}S_iI_j}{\sum_{i=1}^{n}S_i\sum_{j=1}^{n}I_j}$$

This ratio can be interpreted as the ratio of the number of new infections in the age-structured model to the number of new infections in the homogenous model. If $\hat{\beta}(t) > 1$ then age-structured dynamics are fast relative to the homogenous dynamics, and vice versa if $\hat{\beta}(t) < 1$.

# 4 Metapopulation model

We consider a metapopulation model with $P$ identical patches that are assumed to be equally connected, such that $c\%$ of transmission arises from cross-patch contact. The model can be defined as

$$
\begin{aligned}
\frac{dS_p}{dt} &= \mu(1 - v_p) - \beta \sum_{q=1}^{P} C_{p,q} S_p I_q - \mu S_p \\
\frac{dI_p}{dt} &= \beta \sum_{q=1}^{P} C_{p,q} S_p I_q - (\gamma + \mu) I_p \\
\frac{dR_p}{dt} &= \mu v_p + \gamma I_p - \mu R_p
\end{aligned}
$$

where $C_{p,q}$ is the connectivity matrix, which has $1 - c$ along the diagonal and $c/(P-1)$ elsewhere. Analagous to the age-structured case, the next generation matrix is given by

$$K = \frac{\beta}{\gamma + \mu} C \overrightarrow{S(t)},$$

where $\overrightarrow{S(t)}$ is the number of susceptibles in each patch at time $t$. We specify effective immunization coverage to decline in $\rho\%$ of these patches while the others remain at pre-decline coverage.

# 5 Supplementary Figures

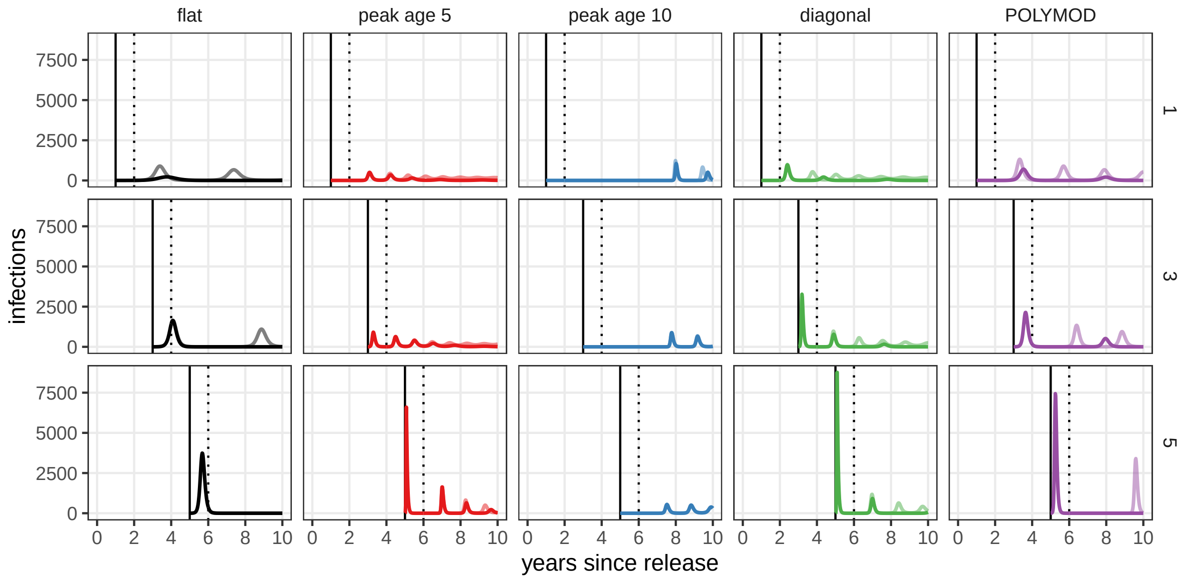


Figure S1: Sensitivity of results to the timing of infection introduction. For each WAIFW matrix (columns, see main text), simulated dynamics when a single infection is introduced into the 5 year old age class 1, 3, or 5 years after the drop in effective immunization coverage (rows). The solid vertical line shows the timing of introduction. The lighter line shows simulation results if effective immunization coverage returns to its pre-decline level 1 year after the drop (shown by dotted line).

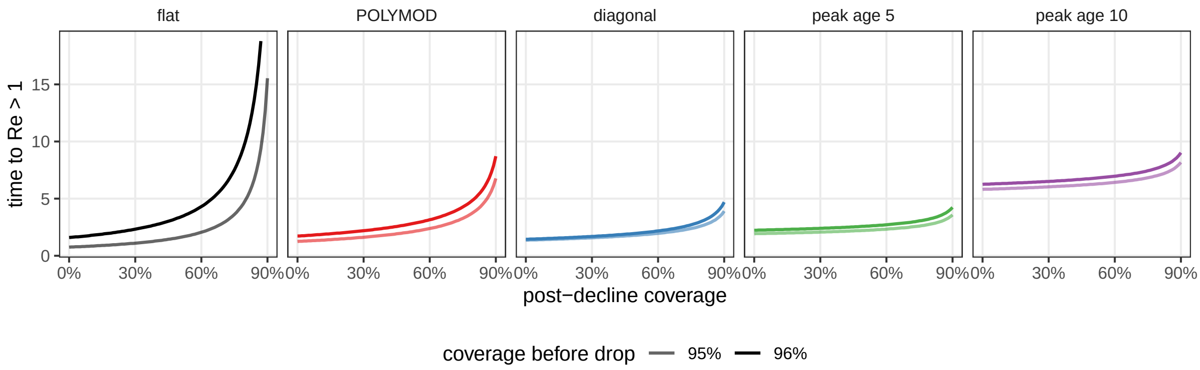


Figure S2: Theoretical honeymoon time predictions, computed as time until Re ¿ 1, under five assumptions about age-structured mixing. Results are shown for a population with $\mu = 1/80$ years$^{-1}$, a pathogen with $\mathcal{R}_0 = 17$, and pre-decline effective immunization coverage of 95% and 96%.

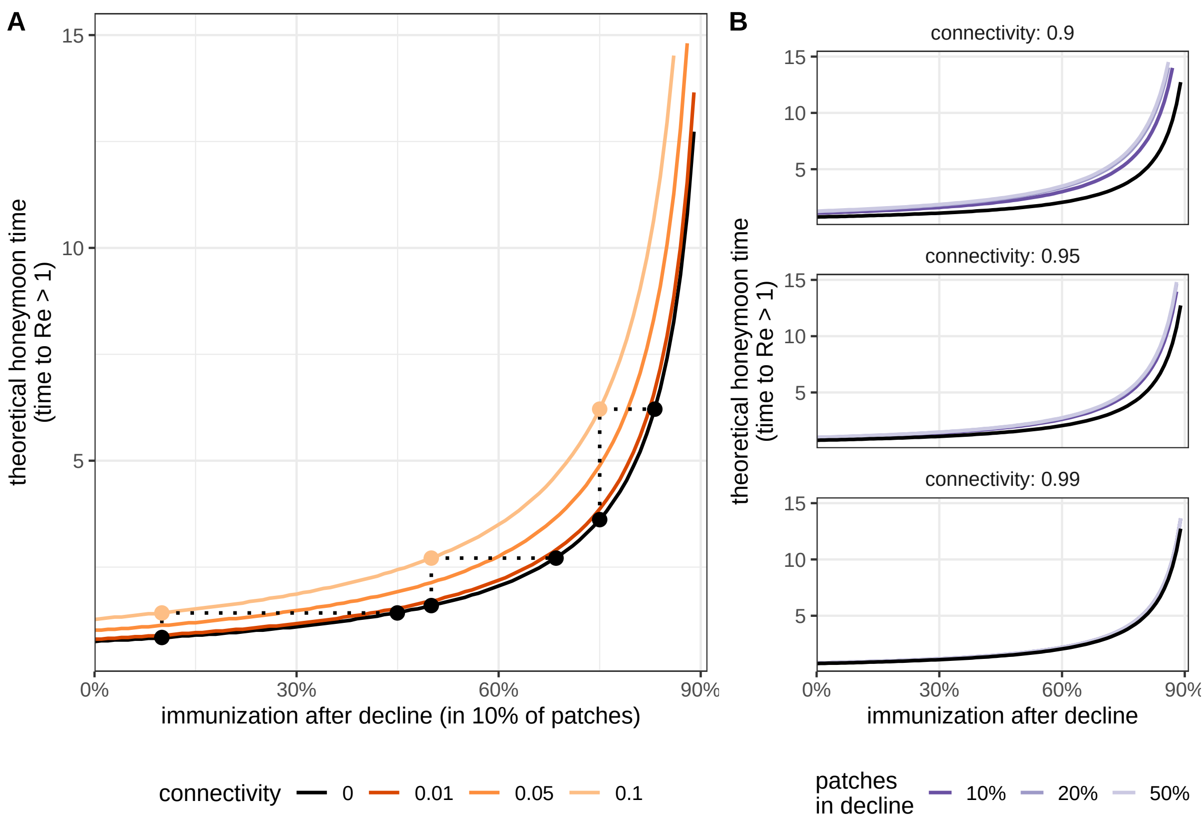


Figure S3: Role of patch connectivity in theoretical honeymoon time. (A) Theoretical honeymoon time, computed as time until Re ¿ 1, by post-decline effective immunization coverage when drops occur in 10% of patches among an equally connected network. Line color shows connectivity between patches (where 0.05 indicates 5% of transmission is shared between patches, or $c$ above), and the points show equivalence between a model with no connectivity and a model where 10% of transmission is shared between patches. (B) Sensitivity of results to the percent of patches in which effective immunization coverage declines, or $\rho$ above. Results are shown for a population with $\mu = 1/80$ years$^{-1}$, a pathogen with $\mathcal{R}_0 = 17$, and pre-decline effective immunization coverage of 95%.